

EXPERIMENT 3

COMPUTATION OF EIGENVALUES AND EIGENVECTORS

Objectives

- a) To write a Scilab program for computing the eigenvalues and eigenvectors of the following matrices

$$\mathbf{X} = \begin{bmatrix} 4 & 3 & 7 \\ 1 & 2 & 7 \\ 2 & 0 & 4 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 & -i & 3+4i \\ i & 2 & 4 \\ 3-4i & 4 & 3 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 2 & -i & 2i \\ i & 4 & 3 \\ -2i & 3 & 5 \end{bmatrix}$$

using the in-built `spec` function.

- b) To verify the results by checking the characteristic equation $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for the computed values.

Theory

In linear algebra, an eigenvector of a square matrix $\mathbf{A}$ is a non-zero vector $\mathbf{v}$ such that when $\mathbf{A}$ multiplies $\mathbf{v}$, the result is a scalar multiple of $\mathbf{v}$. This scalar is known as the *eigenvalue* λ . The relation is expressed as:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Scilab provides the powerful `spec()` function to solve this problem numerically.

When called as the following:

$$[\text{eigenvector}, \text{eigenvalue_matrix}] = \text{spec}(\mathbf{A})$$

it returns a matrix containing eigenvectors as columns and a diagonal matrix containing the corresponding eigenvalues.

For Hermitian matrices (where $\mathbf{A} = \mathbf{A}^\dagger$), the eigenvalues are always real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.

Scilab Code

```
1 // Function to compute and display results
2 function compute_spec(A, name)
3     printf("\nResults for Matrix %s:\n", name);
4     [v, D] = spec(A);
5     lambda = diag(D);
6     disp("Eigenvalues:", lambda');
7     disp("Eigenvectors (along columns):", v);
8
9     // Verification: A*v(:,1) - lambda(1)*v(:,1) should be approx
10    zero
11    verification = A*v(:,1) - lambda(1)*v(:,1);
12    disp("Verification using residual for 1st eigenpair:", clean(
13        verification));
14 endfunction
15
16 // Matrix 1: Real Matrix
17 X = [4, 3, 7; 1, 2, 7; 2, 0, 4];
18
19 // Matrix 2: Complex Hermitian Matrix
20 // Note: %i represents the imaginary unit in Scilab
21 Y = [1, -%i, 3+4*%i; %i, 2, 4; 3-4*%i, 4, 3];
22
23 // Matrix 3: Complex Hermitian Matrix
24 Z = [2, -%i, 2*%i; %i, 4, 3; -2*%i, 3, 5];
25
26 // Execute for the given matrices
27 compute_spec(X, "X");
28 compute_spec(Y, "Y");
29 compute_spec(Z, "Z");
```

Output

Console output of Scilab code:

```
--> exec('D:\Professional\Work\RC\courses\Lab\Sem-04_Math-Phys\expt-03.sce', -1)

Results for Matrix X:
"Eigenvalues:"
8.7276868 + 0.i    0.6361566 - 1.8684093i    0.6361566 + 1.8684093i
"Eigenvectors (along columns):"
0.8096035 + 0.i    0.0303972 + 0.6031288i    0.0303972 - 0.6031288i
0.4766966 + 0.i    -0.7326594 + 0.i            -0.7326594 + 0.i
0.3424946 + 0.i    0.1384051 - 0.2817195i    0.1384051 + 0.2817195i
"Verification using residual for 1st eigenpair:"
0.
0.
0.

Results for Matrix Y:
"Eigenvalues:"
-4.746829    2.3968018    8.3500273
"Eigenvectors (along columns):"
-0.3404771 - 0.5212558i    -0.2734105 + 0.5541666i    -0.3192331 - 0.3586139i
-0.4539695 + 0.0504648i    -0.3645474 - 0.6890355i    -0.4256442 - 0.0502727i
0.6353997 + 0.i            0.1023784 + 0.i            -0.7653665 + 0.i
"Verification using residual for 1st eigenpair:"
0. + 0.i
0. + 0.i
0. + 0.i

Results for Matrix Z:
"Eigenvalues:"
-0.3871996    3.6916109    7.6955887
"Eigenvectors (along columns):"
0.          + 0.6662617i    0.          - 0.7255184i    0.          + 0.1723904i
0.5184148 + 0.i            0.6168052 + 0.i            0.5922816 + 0.i
-0.5360424 + 0.i            -0.3052448 + 0.i            0.7870731 + 0.i
"Verification using residual for 1st eigenpair:"
0. + 0.i
0. + 0.i
0. + 0.i

--> |
```

Results

- The eigenvalues and eigenvectors of the given matrices were computed using a Scilab program.
- The results were verified by checking the characteristic equation for the computed values.